

A realisability model for CIC^- with the full universe hierarchy

Assemblies over a normalisation model, graded by its accessibility ranks

Draft prepared with Claude (Cowork), for Loïc Pujet

September 2026

Abstract

We give a detailed paper proof of the consistency of CIC^- — MLTT with an impredicative sort of propositions closed under $\perp$ and Eq with large elimination, no accessibility — with its whole universe hierarchy, in a metatheory that is plain CIC with impredicative Prop and the large elimination of Acc . The construction has two layers. The first is a truth-agnostic normalisation model on closed terms, in Allen’s PER style, which assigns to every type an accessibility rank. The second is a hierarchy of *assembly codes* built by recursion on those ranks: a code is a normalisation type together with a truth-sensitive refinement, and a proof type is refined by a truth *value*, which is what prevents the cross-level circularity that defeated earlier attempts. Checking the dodgy steps of the earlier sketch forced three changes, recorded in Section 1: layer 1 needs a clause for stuck closed terms; codes must carry their layer-1 shadow; and the decoding of a Π -code must be restricted to functions that respect the equality. We prove the closure, coherence and rank lemmas in full, and identify what is left open.

Contents

1	What checking the sketch changed	1
2	Preliminaries	2
2.1	The object theory	2
2.2	Erased terms and reduction	2
3	Layer 1: a normalisation model on closed terms	3
4	Ranks	3
5	Assembly codes over layer 1	4
6	The interpretation	6
7	The fundamental lemma	6
8	What is proved, and what is not	8

1 What checking the sketch changed

The previous note listed four risky steps. Checking them produced the following, and the reader who knows the earlier drafts should read this section first.

1. **Layer 1 needs stuck terms.** The rank of a type is the rank of its erasure in the normalisation model, so *every* well-typed type must have a layer-1 reducible erasure. This

fails as stated: $x:\text{Prf} \perp \vdash \text{absurd}_{\text{Type}_0}(x) : \text{Type}_0$ has an erasure that is a stuck constant. The repair is the standard one for normalisation models, and it costs nothing here: layer 1 gets a clause for *closed stuck* terms. No open terms, no variables and no renamings are involved — stuck closed terms exist only because of **absurd** — so the closed-term discipline of the earlier audit survives intact (Remark 3.2).

2. **Codes must carry their layer-1 shadow.** Without it, the invariant that makes ranks bound stages is false. The counterexample is $\Pi(x:\text{Prf} \perp).\text{N}$: its semantic elements are the empty function, realised by *any* closed term, whereas the rank argument needs realisers to be layer-1 reducible. Carrying a shadow $\text{sh}(c)$ and demanding, uniformly in every code, that realisers be layer-1 good in $\text{sh}(c)$ makes the invariant true by construction (Definition 5.1, Lemma 5.3).
3. **El of a Π -code keeps only functions respecting $\sim$.** Otherwise $\sim$ is not reflexive on El, and the induction hypothesis at an extended environment cannot be applied, since that needs the extended environment to be related to itself.

With these in place the four risky points go through: the rank bound becomes near-definitional (Lemma 5.2 and the “Stages” bullet in the Π case of Theorem 7.1), the coherence side condition of the Π -code is supplied by the induction hypothesis at a subderivation (the “coh” bullet of the same case), and transport is the $\cong$ -transport, which is the identity at refl (Lemma 5.6).

Sections 2–4 set up layer 1 and the ranks; Sections 5–6 build the codes and the interpretation; Section 7 proves the fundamental lemma; Section 8 lists what is not checked.

2 Preliminaries

2.1 The object theory

CIC^- has: a predicative hierarchy $\text{Type}_0, \text{Type}_1, \dots$ with explicit embeddings $\uparrow$; Π with η ; Σ ; N with large elimination; an impredicative Prop with Prf , closed under $\forall(x:A).p$ for A of any sort; $\perp : \text{Prop}$ with large elimination absurd_T ; and Eq in Prop with transport. We assume the rules are given in the redundant style, so that each rule carries the formation judgements of the types it mentions.

2.2 Erased terms and reduction

Λ is the inductive type of closed untyped terms:

$$\begin{aligned} t ::= & \lambda x.t \mid t \cdot u \mid \langle t, u \rangle \mid \pi_i t \mid 0 \mid S t \mid \text{rec}(t, u, v) \\ & \mid \text{N} \mid \Pi t u \mid \Sigma t u \mid \text{Prop} \mid \text{Prf } t \mid \forall t u \mid \perp \mid \text{Eq } t u v \mid u_n \mid \text{err}. \end{aligned}$$

$\rightsquigarrow$ is weak head reduction, with the usual rules and head congruences; **err** has no reduction rules and $\text{err} \cdot u$, $\pi_i \text{err}$, $\text{rec}(z, s, \text{err})$ do not reduce. Erasure $|-|$ forgets annotations, sends every proof term and every transport to its subject or to $\star$, and sends $\text{absurd}_T(e)$ to **err**. Type formers keep their codomains as functions: $|\Pi(x:A).B| = \Pi |A| (\lambda x. |B|)$.

Definition 2.1 (Stuck terms). t is *stuck* if it is a whnf whose head is **err**: the set of stuck terms is generated by **err** and closed under $(-)\cdot u$, π_i , $\text{rec}(z, s, -)$, $\text{Prf}(-)$, El-like eliminations and so on. A stuck term is closed; no variables are involved.

Lemma 2.2 (Determinism). $\rightsquigarrow$ is deterministic, and $t \Downarrow w$, $t \Downarrow w'$ imply $w = w'$.

Proof. Induction and inversion; the head congruences have disjoint subjects, and stuck terms are irreducible by definition. $\square$

3 Layer 1: a normalisation model on closed terms

Definition 3.1 (The type system). $\text{PER} := \Lambda \rightarrow \Lambda \rightarrow \text{Prop}$. LR_X , for a parameter $X : \Lambda \rightarrow \Lambda \rightarrow \text{PER} \rightarrow \text{Prop}$, is the Prop -valued inductive family with clauses: closure under $\doteq$; closure under expansion on both sides; $\mathbf{N}$ with NatPer ; Prop with PR ; $\text{Prf } p$ with the transparent $\text{PER } \lambda a b. \top$, given $\text{PR } p p'$; $\mathbf{u}$ with $\lambda C C'. \exists P. X C C' P$; Π and Σ with P_A, P_B as constructor arguments; and

$$\frac{N, N' \text{ stuck}}{\text{LR}_X N N' \text{NePer}} \quad \text{NePer } a b := a \text{ and } b \text{ are stuck or } \top - \text{ see below.}$$

$\text{NatPer } a b := (\exists k. a \Downarrow \bar{k} \wedge b \Downarrow \bar{k}) \vee (a, b \text{ stuck})$, and every other PER above is likewise closed under stuck terms: $\Pi\text{-Per}$ and $\Sigma\text{-Per}$ are as before but with stuck terms adjoined, and $\text{NePer } a b$ holds iff a and b are stuck. PR is the shape-only relation on propositions, with a stuck clause. Finally $\tau_0 := \text{LR}_{\lambda \dots \perp}$ and $\tau_{n+1} := \text{LR}_{\tau_n}$, and $A \approx_n A'$ abbreviates $\exists P. \tau_n A A' P$.

Remark 3.2 (Still closed, still no renamings). The stuck clause quantifies over closed stuck terms, which exist because $|\text{absurd}_T(e)| = \text{err}$. No variable ever appears in a PER , no term is ever reduced under a binder, and no Kripke index or renaming lemma is needed. The audit of the earlier note therefore still applies verbatim; what changes is only that the PER s are inhabited by junk in impossible contexts, which is exactly what makes Lemma 3.4 true.

Lemma 3.3 (Basic theory of τ). *By one simultaneous induction on derivations, for all n : τ_n is symmetric and transitive; each P is a PER closed under expansion and containing the stuck terms; τ_n is functional, i.e. $\tau_n A A' P$ and $\tau_n A A'' Q$ imply $P \doteq Q$; the codomain families satisfy $P_B a a' \doteq P_B b b'$ for P_A -related arguments; $\tau_m \subseteq \tau_n$ for $m \leq n$; and inversion by shape holds, with the whnfs determined by Lemma 2.2.*

Proof. This is the standard Allen-style bundle, and the parts cannot be separated: symmetry at Π needs index irrelevance, which needs functionality at the codomain, which needs symmetry and transitivity at the domain. The stuck clause adds no difficulty: stuck terms have no reducts, so the expansion clause cannot interfere, and NePer is trivially a PER . Cumulativity is by an outer induction on n , using monotonicity of LR in its parameter, which holds because every clause is positive in X . $\square$

Lemma 3.4 (Fundamental lemma of layer 1). *Let $\sigma \approx_\Gamma \sigma'$ mean that the substitutions of closed terms are pointwise related by the PER s of the context's types. If $\Gamma \vdash A : \text{Type}_k$ then $|A|\sigma \approx_k |A|\sigma'$; and if $\Gamma \vdash t : A$ then $|t|\sigma$ and $|t|\sigma'$ are related by the PER of $|A|\sigma$.*

Proof sketch. Induction on the derivation; this is the closed-substitution form of the POPL'23 normalisation model, so we only comment on the clauses that differ. **absurd**: the erasure is err , which is stuck and hence in every PER by Lemma 3.3; this is the case that fails without the stuck clause. **N-elimination**: the scrutinee is in NatPer , so it either converges to a numeral — induct — or is stuck, in which case the whole recursor is stuck and we conclude by the stuck clause. *Proof terms and transports*: erasures are $\star$ or the subject, and the Prf - PER is transparent, so these cases are immediate. This is also where proof irrelevance of the erasure is used: a proof variable never occurs in an erasure, so a junk realiser in a proof slot does no harm. *Impredicative $\forall$* : PR is shape-only and takes the domain PER as a constructor argument, so the induction hypothesis supplies it; no level bookkeeping is involved. $\square$

4 Ranks

Definition 4.1 (Ord). Ord is the inductive type generated by $\text{sup} : (\Lambda \times \Lambda \rightarrow \text{Ord}) \rightarrow \text{Ord}$ (so $\text{zero} := \text{sup}$ of the constantly-smaller family is definable), with the usual $\leq$ and $<$. It lives in Set and $<$ is well founded.

Definition 4.2 (The order $\prec$). $C \prec A$ iff $A \Downarrow w$ and either

- $w = \Pi A_0 B_0$ or $\Sigma A_0 B_0$ and $C = A_0$; or
- $w = \Pi A_0 B_0$ or $\Sigma A_0 B_0$ and $C = B_0 \cdot u$ for some u with $P u u$, where $\exists n. \tau_n A_0 A_0 P$; or
- $w = u_m$ and $C \approx_m C$.

Lemma 4.3 (Accessibility). *If $A \approx_n A$ then $\text{Acc} \prec A$.*

Proof. By induction on n , then on the derivation of $\tau_n A A P$. The $\doteq$ and expansion clauses leave the whnf unchanged (Lemma 2.2), so the predecessors are the same. **N**, **Prop**, **Prf** and the stuck clause have no predecessors, so **Acc** holds by its constructor with a vacuous premise. At Π and Σ the predecessors are A_0 , accessible by the inner induction hypothesis, and the $B_0 \cdot u$ for u in *some* PER of A_0 ; by functionality (Lemma 3.3) that PER is $\doteq$ to the one in the derivation, so the inner induction hypothesis applies to exactly these. At u_m the predecessors are all level- m types, accessible by the *outer* induction hypothesis, since $m < n$. $\square$

Definition 4.4 (Rank). $\text{rk} : \forall A. \text{Acc} \prec A \rightarrow \text{Ord}$ by Acc_rect into **Set**: $\text{rk}(A, h) := \sup(\lambda q. \text{if } q \prec A \text{ then } \text{rk}(q, h_q) + 1 \text{ else zero})$, the family being indexed by $\Lambda \times \Lambda$ as required by **Ord** (the second component is ignored).

Lemma 4.5 (Properties of rk). (R1) *rk does not depend on the accessibility proof.*

(R2) *If $C \prec A$ then $\text{rk}(C) < \text{rk}(A)$. In particular the domain and all codomain instances of a Π - or Σ -type have rank below it.*

(R3) *$\beta_m := \text{rk}(u_m)$ strictly bounds the ranks of all level- m types, and $\beta_m < \beta_{m'}$ for $m < m'$.*

(R4) *If $A \approx_n A'$ then $\text{rk}(A) \leq \text{rk}(A')$ and conversely.*

Proof. (R1) By Acc -induction: the value depends only on $\prec$, which is a fixed relation. (R2) By construction of the $\sup$. (R3) The first part is (R2) with the universe clause of $\prec$; the second holds because u_m is itself a level- m' type for $m < m'$, so $\beta_m < \beta_{m'}$ by (R2). (R4) By Acc -induction. Inversion (Lemma 3.3) makes the whnfs of A, A' have the same shape, with $A_0 \approx A'_0$ and $B_0 \cdot u \approx B'_0 \cdot u$ for the same u 's — the domain PERs coincide by functionality — so every predecessor of A has a $\approx$ -partner among the predecessors of A' of no greater rank, whence $\text{rk}(A) \leq \text{rk}(A')$; symmetrically for the converse. $\square$

5 Assembly codes over layer 1

A code is a layer-1 type together with a truth-sensitive refinement. Writing $\text{sh}(c)$ for the underlying layer-1 type and $P_{\text{sh}(c)}$ for its PER, the design is fixed by two requirements: a proof type must be refined by a truth *value*, so that the hierarchy never mentions propositions-as-syntax; and the decoding must force realisers to be layer-1 good, so that ranks bound stages.

Definition 5.1 (The hierarchy). By well-founded recursion on $\alpha : \text{Ord}$ we define $\mathcal{U}_\alpha : \text{Type}_1$, a map $\text{sh} : \mathcal{U}_\alpha \rightarrow \Lambda$, a decoding $\text{El}_\alpha : \mathcal{U}_\alpha \rightarrow \Lambda \rightarrow \text{Type}_1$ and an equality $\sim_\alpha$ on $\sum_u \text{El}_\alpha(c) u$.

An element of $\mathcal{U}_\alpha$ is a pair of a term T with $T \approx_n T$ for some n and $\text{rk}(T) < \alpha$, together with refinement data given by cases on the whnf of T :

whnf of T	refinement data	$\text{El}_\alpha(c) u$
N	none	$\{k \mid u \Downarrow \bar{k}\} \times \ P_T u u\ $
Prop	none	$\text{Prop} \times \ P_T u u\ $
Prf p	$H : \text{Prop}$	$H \times \ P_T u u\ $
$\mathbf{u}_m$	none	$\{c' \in \mathcal{U}_{\beta_m} \mid \text{sh}(c') \approx_m u\}$
$\Pi A_0 B_0$	$a \in \mathcal{U}_{\text{rk}(A_0)+1}$ and $b : \prod_{u'} \text{El}(a) u' \rightarrow \mathcal{U}_{<\alpha}$	$\{f \mid f \text{ respects } \sim \text{ and } f u' x \in \text{El}(b u' x)(u \cdot u')\}$ $\times \ P_T u u\ $
$\Sigma A_0 B_0$	as for Π	$(\sum_{x:\text{El}(a)(\pi_1 u)} \text{El}(b(\pi_1 u) x)(\pi_2 u))$ $\times \ P_T u u\ $
stuck	none	$\emptyset$

subject to the coherence conditions $\text{sh}(a) \approx A_0$, $\text{sh}(b u' x) \approx B_0 \cdot u'$, and

$$\text{coh} : (u', x) \sim_a (u'', x'') \implies b u' x \cong b u'' x''.$$

The equality $\sim_\alpha$ requires in every case that the two realisers be P_T -related, and in addition: the same numeral for **N**; nothing for **Prf**; $H \leftrightarrow H'$ for **Prop**; $\cong$ for $\mathbf{u}_m$; pointwise on $\sim$ -related arguments for Π ; componentwise for Σ .

Lemma 5.2 (The recursion is well founded). *Definition 5.1 is a legitimate well-founded recursion on α : every appeal to $\mathcal{U}$, El or $\sim$ on the right-hand side is at a stage $< \alpha$.*

Proof. The Π and Σ cases mention $\mathcal{U}_{\text{rk}(A_0)+1}$ and $\mathcal{U}_{<\alpha}$, and $\text{rk}(A_0) < \text{rk}(T) < \alpha$ by (R2); the values $b u' x$ are required to lie in $\mathcal{U}_{\text{rk}(B_0 \cdot u')+1}$, and $\text{rk}(B_0 \cdot u') < \text{rk}(T)$, again by (R2), *provided* u' is P_{A_0} -related to itself, which the decoding of a guarantees by its $\|P_{\text{sh}(a)} u' u'\|$ component together with $\text{sh}(a) \approx A_0$ and Lemma 3.3. The universe case mentions $\mathcal{U}_{\beta_m}$ with $\beta_m < \text{rk}(\mathbf{u}_m) + 1 \leq \alpha$ by (R3). No other case is recursive. Note that no induction-recursion is involved: at stage α all earlier decodings are already available, and nothing is defined simultaneously with its own decoding. $\square$

Lemma 5.3 (Soundness of realisers). *If $x \in \text{El}(c) u$ then $P_{\text{sh}(c)} u u$; and if $(u, x) \sim_c (u', x')$ then $P_{\text{sh}(c)} u u'$.*

Proof. Both hold by construction: the $\|P_T u u\|$ component is part of every non-universe case, the universe case gives $\text{sh}(c') \approx_m u$ whence $P_{\mathbf{u}_m} u u$, and the stuck case is vacuous. The second claim is the first clause of $\sim$. $\square$

Lemma 5.4 ($\sim$ is an equivalence). *For every c , $\sim_c$ is reflexive, symmetric and transitive on $\sum_u \text{El}(c) u$.*

Proof. Induction on the stage. Symmetry and transitivity are componentwise, using Lemma 3.3 for the realiser component. Reflexivity is the only interesting case: at Π it is exactly the “respects $\sim$ ” condition, which is why that condition is part of the decoding; at Σ it follows from reflexivity at the components; at $\mathbf{u}_m$ it is reflexivity of $\cong$ (Lemma 5.6). $\square$

Definition 5.5 (Isomorphism). $c \cong c'$ iff $\text{sh}(c) \approx \text{sh}(c')$ — so their whnfs have the same shape — and the refinements correspond: $H \leftrightarrow H'$ for **Prf**; $a \cong a'$ and $b u x \cong b' u' x'$ whenever (u, x) and (u', x') correspond, for Π and Σ ; nothing for the other cases.

Lemma 5.6 (Isomorphism and transport). *$\cong$ is an equivalence relation; if $c \cong c'$ then there is a family of bijections $\theta_u : \text{El}(c) u \rightarrow \text{El}(c') u$, natural in u and preserving $\sim$; and for the reflexivity witness $c \cong c$ the bijection is the identity. Moreover $c \cong c'$ implies $\mathcal{U}_\alpha \ni c \Leftrightarrow c' \in \mathcal{U}_\alpha$, by (R4).*

Proof. Induction on the stage. At **Prf** the bijection is transport along $H \leftrightarrow H'$ between two propositions; at **N**, **Prop**, **u** and **stuck** it is the identity modulo the realiser side conditions, which transfer by Lemma 3.3; at **II** it is conjugation by the bijections at domain and codomain, which is where **coh** is used, and preservation of $\sim$ follows from naturality. That the reflexivity witness induces the identity is proved by the same induction, each clause being built from identities. $\square$

Lemma 5.7 (Expansion closure of the decoding). *If $u \rightsquigarrow^* u_1$ then $\text{El}(c)u$ and $\text{El}(c)u_1$ are naturally isomorphic, compatibly with $\sim$.*

Proof. Induction on the stage. At **N** both sides are $\{k \mid u \Downarrow \bar{k}\}$, unchanged because $\Downarrow$ is closed under expansion; at **Prf** and **Prop** the decoding does not mention u except through $\|P_T u u\|$, which is expansion-closed by Lemma 3.3; at **u_m** the condition $\text{sh}(c') \approx_m u$ is expansion-closed for the same reason; at **II**, $u \cdot u' \rightsquigarrow^* u_1 \cdot u'$ and one appeals to the induction hypothesis at the codomain, and similarly for Σ through the projections. $\square$

6 The interpretation

Definition 6.1 (Environments). A semantic environment ρ for Γ assigns to each $(x:A) \in \Gamma$ a pair (u, x) with $x \in \text{El}(\llbracket A \rrbracket \rho)u$; $|\rho|$ is the induced substitution of realisers. Two environments are related, $\rho \approx_\Gamma \rho'$, when for each variable the pairs are $\sim$ -related after transporting along the isomorphism $\llbracket A \rrbracket \rho \cong \llbracket A \rrbracket \rho'$ of Lemma 5.6.

Definition 6.2 (Interpretation). $\llbracket t \rrbracket \rho$ is defined by recursion on raw syntax, partially: an expression is interpreted only when its constituents are and the side conditions of Definition 5.1 hold. Types are interpreted by codes, terms by elements, propositions by truth values:

$$\begin{aligned} \llbracket \mathbf{N} \rrbracket \rho &= \text{the N-code}, & \llbracket \mathbf{Type}_k \rrbracket \rho &= \text{the } u_k\text{-code}, \\ \llbracket \mathbf{Prop} \rrbracket \rho &= \text{the Prop-code}, & \llbracket \perp \rrbracket \rho &= \text{False}, \\ \llbracket \mathbf{Prf } p \rrbracket \rho &= \text{the Prf-code with } H := \llbracket p \rrbracket \rho, & \llbracket \uparrow A \rrbracket \rho &= \llbracket A \rrbracket \rho, \end{aligned}$$

and, writing $\rho[u, x]$ for $(\rho, x \mapsto (u, x))$,

$$\begin{aligned} \llbracket \Pi(x:A).B \rrbracket \rho &= \text{the } \Pi\text{-code with } a := \llbracket A \rrbracket \rho \text{ and } b := \lambda u x. \llbracket B \rrbracket \rho[u, x], \\ \llbracket \forall(x:A).p \rrbracket \rho &= \forall u x. x \in \text{El}(\llbracket A \rrbracket \rho)u \rightarrow \llbracket p \rrbracket \rho[u, x], \\ \llbracket \text{Eq } A t u \rrbracket \rho &= ((|t||\rho|, \llbracket t \rrbracket \rho) \sim_{\llbracket A \rrbracket \rho} (|u||\rho|, \llbracket u \rrbracket \rho)), \end{aligned}$$

with the evident clauses for λ , application, pairs, projections, numerals and the recursor, and with every proof term interpreted by the unique element of its truth value.

Lemma 6.3 (Semantic substitution). $\llbracket t[a/x] \rrbracket \rho = \llbracket t \rrbracket (\rho, x \mapsto (|a||\rho|, \llbracket a \rrbracket \rho))$, and $|t[a/x]| \rho| = |t|(|\rho|, x \mapsto |a||\rho|)$.

Proof. The second statement is the erasure-substitution lemma, by induction on raw syntax; the first is by induction on t , every clause of Definition 6.2 being compositional. $\square$

7 The fundamental lemma

Theorem 7.1. *Let $\rho \approx_\Gamma \rho'$. Then:*

- (a) *if $\Gamma \vdash A : \mathbf{Type}_k$ then $\llbracket A \rrbracket \rho$ is a defined code with $\text{sh}(\llbracket A \rrbracket \rho) = |A||\rho|$, of rank $< \beta_k$, hence $\llbracket A \rrbracket \rho \in \mathcal{U}_{\beta_k}$; and $\llbracket A \rrbracket \rho \cong \llbracket A \rrbracket \rho'$;*
- (b) *if $\Gamma \vdash t : A$ then $\llbracket t \rrbracket \rho \in \text{El}(\llbracket A \rrbracket \rho)(|t||\rho|)$ and $\llbracket t \rrbracket \rho \sim \llbracket t \rrbracket \rho'$ after transport;*

(c) if $\Gamma \vdash A \equiv B$ then $\llbracket A \rrbracket \rho \cong \llbracket B \rrbracket \rho'$, and if $\Gamma \vdash t \equiv u : A$ then $\llbracket t \rrbracket \rho \sim \llbracket u \rrbracket \rho'$ after transport.

Proof. By induction on the derivation. Throughout, Lemma 5.3 gives $|\rho| \approx_\Gamma |\rho'|$, so Lemma 3.4 applies and supplies, for every type of the derivation, that its erasure is layer-1 reducible and that the two erasures are $\approx$ -related. This is what makes (a) meaningful: the shadow exists. We treat the cases.

Variables and conversion. Immediate from the definitions, using Lemma 5.6 to transport along $\cong$ in the conversion case.

(a) for Π — the risky point about coh. Put $a := \llbracket A \rrbracket \rho$ and $b := \lambda u x. \llbracket B \rrbracket (\rho, (u, x))$. Three things must be checked.

- *b is defined.* For $x \in \text{El}(a) u$, the extended environment is related to itself: the realiser component holds by Lemma 5.3 and the semantic component by reflexivity of $\sim$ (Lemma 5.4). So the induction hypothesis (a) for B applies and $b u x$ is a defined code, with shadow $|B|(|\rho|, u)$, which is $\approx$ to $(\lambda x. |B|(|\rho|)) \cdot u$ by expansion closure.
- *Stages.* By Lemma 5.3 the realiser u satisfies $P_{\text{sh}(a)} u u$, and $\text{sh}(a) = |A|(|\rho|)$, so $(\lambda x. |B|(|\rho|)) \cdot u$ is a $\prec$ -predecessor of $|\Pi(x:A).B|(|\rho|)$. By (R2) and (R4) its rank is strictly smaller, so $b u x$ lies at a stage below the rank of the shadow, which is what Definition 5.1 requires; the same argument with the first clause of $\prec$ handles a .
- *coh.* If $(u, x) \sim_a (u', x')$ then the two extended environments are related, by Lemma 5.3 for the realiser component; so the induction hypothesis (a) for B , second part, gives $b u x \cong b u' x'$. There is no circularity: this is the induction hypothesis at the *subderivation* B , which is available before the code for $\Pi(x:A).B$ is formed.

Finally $\llbracket \Pi(x:A).B \rrbracket \rho \cong \llbracket \Pi(x:A).B \rrbracket \rho'$ follows from the induction hypotheses for A and B and Definition 5.5. The rank of the shadow is $< \beta_k$ by (R3), since the shadow is a level- k type by Lemma 3.4. Σ is identical.

(b) for λ and application. For λ , set $f := \lambda u x. \llbracket t \rrbracket (\rho, (u, x))$. Membership requires $f u x \in \text{El}(b u x)((\lambda x. |t|(|\rho|)) \cdot u)$; the induction hypothesis gives membership at the realiser $|t|(|\rho|, u)$, and the two realisers differ by one β -step, so Lemma 5.7 applies. That f respects $\sim$ is the second part of the induction hypothesis at related extended environments. The layer-1 side condition $\|P_T t t\|$ is Lemma 3.4. Application is the defining clause of El at a Π -code, together with Lemma 6.3 to match $\llbracket B[u/x] \rrbracket \rho$ with b at the argument.

η . $\llbracket \lambda x. t \cdot x \rrbracket \rho$ and $\llbracket t \rrbracket \rho$ are the same semantic function, by η in the metatheory; the realisers differ by an η -expansion, which layer 1 relates because its Π -PER is extensional, and El does not distinguish them by Lemma 5.7.

N and its large elimination. $\text{El}(\text{nat}) u$ forces $u \Downarrow \bar{k}$; note that the stuck inhabitants of NatPer are *not* semantic elements, which is exactly why layer 1 may be truth-agnostic while layer 2 is not. For the recursor, induct on k : the computation rules are matched by Lemma 5.7, and the motive instances are $\llbracket C \rrbracket (\rho, (\bar{k}, k))$, defined and of bounded stage by the induction hypothesis (a) at each numeral.

Universes and embeddings. $\llbracket \text{Type}_k \rrbracket \rho$ is the u_k -code; it is an element of $\text{El}(\llbracket \text{Type}_{k+1} \rrbracket \rho)(u_k)$ because $\text{sh}(u_k\text{-code}) = u_k \approx_{k+1} u_k$, and its rank β_k is $< \beta_{k+1}$ by (R3). For $\uparrow A$, the erasure is unchanged and the code is the same; membership at the next universe is (R3) again. The decoding equation $\text{El}(\uparrow A) \equiv \text{El}(A)$ holds because both sides are interpreted by the same code.

Prop, Prf, $\forall$, $\perp$. $\llbracket p \rrbracket \rho$ is a truth value, an element of $\text{El}(\text{Prop-code})(|p||\rho|)$ — the side condition $\llbracket \text{PR} \rrbracket$ is Lemma 3.4. For $\forall$ -introduction, the induction hypothesis at every (u, x) with $x \in \text{El}(\llbracket A \rrbracket \rho) u$ gives precisely the metatheoretic $\forall$ that defines $\llbracket \forall(x:A).p \rrbracket \rho$; this is a **Prop** by impredicativity of the metatheoretic **Prop**, whatever the level of A , and no stratification is involved because $\text{El}(\llbracket A \rrbracket \rho)$ is a completed object. Elimination is instantiation, with Lemma 6.3. Proof terms are interpreted by the unique element of a true proposition; all proofs of a given proposition are $\sim$ -related, so proof irrelevance is validated. For **absurd_T**(e): the induction hypothesis gives $\llbracket e \rrbracket \rho \in \text{El}(\text{Prf-code with } H = \text{False})(|e||\rho|) = \text{False}$, so no such ρ exists and the case is vacuous.

Eq and transport — the fourth risky point. $\llbracket \text{Eq } A \text{ } t \text{ } u \rrbracket \rho$ is the proposition $(|t||\rho|, \llbracket t \rrbracket \rho) \sim (|u||\rho|, \llbracket u \rrbracket \rho)$. **refl** is validated by reflexivity of $\sim$ (Lemma 5.4). For transport along $e : \text{Eq } A \text{ } t \text{ } u$ in a family B : the induction hypothesis (a) for B gives **coh**, so from $\llbracket t \rrbracket \rho \sim \llbracket u \rrbracket \rho$ we get $\llbracket B[t/x] \rrbracket \rho \cong \llbracket B[u/x] \rrbracket \rho$, and the transport of Lemma 5.6 maps El of one to El of the other, naturally in the realiser — which matters because the erasure of a transport is its subject, so the realiser must not change. The computation rule at **refl** holds because the transport induced by the reflexivity witness is the identity, again Lemma 5.6.

(c) Conversion. By induction on the conversion derivation. Reflexivity, symmetry, transitivity and congruence follow from Lemma 5.6 and compositionality. Each computation rule is discharged by Lemma 6.3 together with Lemma 5.7: the two sides have the same semantic value and realisers differing by weak head reduction. Crucially, coherence at a universe — the statement that broke the previous design — is now part of this induction rather than a property of layer 1: $\cong$ compares refinements, and two convertible propositions have equal truth values because they have the same interpretation. $\square$

Corollary 7.2 (Consistency). *There is no closed term $\vdash e : \text{Prf } \perp$.*

Proof. The empty environment is related to itself, so Theorem 7.1(b) would give an element of $\text{El}(\llbracket \text{Prf } \perp \rrbracket)(|e|)$, whose first component is an element of **False**. $\square$

Corollary 7.3 (Canonicity for $\mathbb{N}$). *If $\vdash t : \mathbb{N}$ then $|t| \Downarrow \bar{k}$ for some k .*

Proof. $\llbracket t \rrbracket \in \text{El}(\text{nat})(|t|)$, whose first component is a k with $|t| \Downarrow \bar{k}$. $\square$

8 What is proved, and what is not

Metatheoretic budget. Impredicative **Prop** (truth values, the $\llbracket \cdot \rrbracket$ side conditions, and the interpretation of $\forall$); **Prop**-valued inductive families (layer 1); **Ord** in **Set**, since every **sup** is indexed by $\Lambda \times \Lambda$; the large elimination of **Acc**, twice — once for **rk**, once for the hierarchy of Definition 5.1; and one universe above **Set** for the codes. Nothing is stratified by the object level except β_k , which is a definable sequence of ordinals, so the theorem is uniform in k : a single statement covers the whole hierarchy. No axioms are used; every notion is closed under $\cong$, $\doteq$ or $\leftrightarrow$ by construction, so neither functional nor propositional extensionality is needed.

The four risky points, resolved. The rank bound is now near-definitional (Lemma 5.2 and the Π case of Theorem 7.1), because a code carries its shadow and $\mathcal{U}_\alpha$ is defined by a condition on the shadow's rank. **coh** is supplied by the induction hypothesis at a subderivation, so there is no circularity. Transport is the $\cong$ -transport, which is the identity at **refl**. Coherence at a universe is part of the conversion induction, not a property of layer 1.

What I have not verified.

1. **Lemma 3.4 in detail.** It is the closed-substitution form of a known result, but “known” is doing work here: the stuck clause has to be threaded through every PER, and the interaction of the stuck clause with the universe clause deserves a careful pass.
2. **Definedness bookkeeping.** Definition 6.2 is a partial function on raw syntax and Theorem 7.1 asserts definedness; I have checked the side conditions case by case but not the standard coherence argument that the interpretation is independent of the derivation.
3. **Lemma 3.3** is stated as a bundle and proved by the usual simultaneous induction; I have not written that induction out, and it is the single largest piece of routine work in the development.
4. **W -types**, which CIC^- may include, are not treated. They would add a clause to layer 1, a code with an inductively defined decoding, and a $\prec$ -clause; I expect no essential difficulty but have not checked the rank argument for them.
5. **(R4)** is stated as mutual $\leq$ rather than equality of Brouwer trees, which is all that is used; if a formalisation wants equality it will need a quotient or a different ordinal representation.

A remark on strength. The construction uses Acc ’s large elimination in two places and impredicative Prop throughout; CIC^- has the latter but not the former, so the Gödelian sanity check of the earlier notes still passes, and the gap is still concentrated in Acc .